

Week 3 - Physics of the Atmosphere: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: $R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_p = 1004 \text{ J kg}^{-1} \text{ K}^{-1}$, latent heat of vaporisation $L_v \approx 2.5 \times 10^6 \text{ J kg}^{-1}$, Wien constant $b = 2.898 \times 10^{-3} \text{ m K}$.

YOU SHOULD TRY Question 1: Using the ideal-gas equation

Air has pressure 985 hPa and temperature 288 K. Calculate its density using

$$p = \rho R_d T.$$

Comment briefly on whether the result has a sensible atmospheric magnitude.

IN CLASS Question 2: A rising parcel expands and cools adiabatically

A sealed flexible dry-air parcel initially has volume $V_1 = 1.00 \text{ m}^3$ at $p_1 = 1000 \text{ hPa}$ and $T_1 = 290 \text{ K}$. It rises to a level where $p_2 = 850 \text{ hPa}$. Assume the motion is dry adiabatic and use

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{R_d/c_p}, \quad R_d/c_p = 0.286.$$

1. Calculate the parcel temperature T_2 .
2. Use the ideal-gas relation for a fixed amount of gas, $pV/T = \text{constant}$, to calculate V_2 .
3. Calculate the parcel density at the start and at the upper level.
4. Explain physically why the parcel both expands and cools as it rises.

YOU SHOULD TRY Question 3: Dalton's law in the atmosphere

Dalton's law of partial pressures states that, for an ideal-gas mixture, the total pressure is the sum of the partial pressures of the individual gases:

$$p = \sum_i p_i.$$

For an ideal-gas mixture, the partial pressure of constituent i is its mole fraction x_i multiplied by the total pressure:

$$p_i = x_i p.$$

At a location where the total pressure is 1000 hPa, suppose dry air contains 78.08% N_2 and 20.95% O_2 by mole fraction.

1. Estimate the partial pressures of N_2 and O_2 .
2. Check that the two values account for almost all of the total pressure, and explain why they do not add to exactly 1000 hPa.
3. Explain why Dalton's law is useful when water vapour is added to dry air.

YOU SHOULD TRY Question 4: First-law thinking for a rising parcel

Use the sign convention

$$dU = \delta Q - p dV,$$

where $p dV$ is work done by the parcel on its surroundings.

1. What does “adiabatic” mean?
2. A parcel rises, expands and does work. If the motion is adiabatic, what happens to its internal energy and temperature?
3. Why can adiabatic motion be a useful approximation for an air parcel?

EXTRA CHALLENGE Question 5: Sensible heating of air

How much energy is required to raise the temperature of 1000 kg of dry air by 5.0 K at constant pressure? Use $c_p = 1004 \text{ J kg}^{-1} \text{ K}^{-1}$.

YOU SHOULD TRY Question 6: Latent heat is a large energy reservoir

Compare the energy in Question 5 with the energy released when 10 kg of water vapour condenses. Use $L_v = 2.5 \times 10^6 \text{ J kg}^{-1}$.

Modelling connection.

This stability problem connects directly to dispersion modelling: the vertical temperature structure controls whether turbulent motions are suppressed or enhanced. Students who also take EART22001 will encounter the same physics in a Gaussian-plume model, but no knowledge of that unit is assumed here. The fuller treatment of vertical profiles follows in Week 4.

IN CLASS Question 7: Potential temperature and atmospheric stability

At three heights the measured pressure and temperature are:

Height (m)	p (hPa)	T (K)
0	1000	290
500	945	287
1000	890	284

Calculate potential temperature at each level using

$$\theta = T \left(\frac{1000 \text{ hPa}}{p} \right)^{0.286}.$$

Then classify the layer as statically stable, neutral or unstable. Explain what this implies qualitatively for the vertical spread of a passive pollutant plume.

EXTRA CHALLENGE Question 8: Sunlight and terrestrial radiation

Use Wien's law,

$$\lambda_{\max} T = b,$$

to calculate the peak-emission wavelength for

1. the Sun, approximated by $T = 5778 \text{ K}$;
2. the Earth's surface, approximated by $T = 288 \text{ K}$.

State the broad part of the electromagnetic spectrum in which each peak lies.

EXTRA CHALLENGE Question 9: From the universal gas constant to R_d

The universal gas constant is $R^* = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ and the molar mass of dry air is approximately $M_d = 28.97 \times 10^{-3} \text{ kg mol}^{-1}$. Show that the specific gas constant for dry air is close to $287 \text{ J kg}^{-1} \text{ K}^{-1}$.